

Week 9 Book exercise 6.5)

Assume that $k_0(\underline{x}, \underline{x}') = \phi(\underline{x})^T \phi(\underline{x}')$ is a valid kernel.
Then

$$\begin{aligned} k(\underline{x}, \underline{x}') &\stackrel{\text{def.}}{=} c k_0(\underline{x}, \underline{x}') = c (\phi(\underline{x})^T \phi(\underline{x}')) \\ &= (\sqrt{c} \phi(\underline{x}))^T (\sqrt{c} \phi(\underline{x}')), \end{aligned}$$

and by analogy

$$\begin{aligned} k(\underline{x}, \underline{x}') &\stackrel{\text{def.}}{=} f(\underline{x}) k_0(\underline{x}, \underline{x}') f(\underline{x}') \\ &= f(\underline{x}) (\phi(\underline{x})^T \phi(\underline{x}')) f(\underline{x}') \\ &= (f(\underline{x}) \phi(\underline{x}))^T (f(\underline{x}') \phi(\underline{x}')). \end{aligned}$$

Thus (6.13) and (6.14) are valid kernels.

Book exercise 6.6)

By assumption, we have

$$q(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$$

Thus

$$k(x, x') = a_n k_1(x, x')^n + a_{n-1} k_1(x, x')^{n-1} + \dots + a_1 k_1(x, x') + a_0$$

By (6.18), (6.17), and (6.13) it follows that (6.15) is a valid kernel.

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To prove (6.17), note that the Gram matrices K_1 and K_2 are positive semidefinite (as we assume $k_1(x, x')$ and $k_2(x, x')$ to be valid kernels) - and it is a necessary and sufficient condition cf. section 6.2, page 28,

Noting that $K = K_1 + K_2$ also is positive semidefinite (you can find many proofs of this online) it follows that

$$K(x, x') = k_1(x, x') + k_2(x, x')$$

is a valid kernel.

To prove (6.18), note that

$$K(x, x') = k_1(x, x') k_2(x, x')$$

$$= (\phi^{(1)}(x)^T \phi^{(2)}(x)) (\phi^{(1)}(x')^T \phi^{(2)}(x'))$$

$$= \left(\sum_{i=1}^M \phi_i^{(1)}(x) \phi_i^{(1)}(x') \right) \left(\sum_{j=1}^N \phi_j^{(2)}(x) \phi_j^{(2)}(x') \right)$$

$$= \sum_{i=1}^M \sum_{j=1}^N (\phi_i^{(1)}(x) \phi_i^{(2)}(x)) (\phi_j^{(1)}(x') \phi_j^{(2)}(x'))$$

$$= \sum_{k=1}^{M \times N} \phi_k(x) \phi_k(x')$$

see (*) below. ϕ_k is a vector

$$= \phi(\underline{x})^T \phi(\underline{x}')$$

where $\phi_j^{(1)}$ denotes the $j=1, \dots, M$ entry on the map function for kernel $k_1(\underline{x}, \underline{x}')$ and ϕ_j denotes the $j=1, \dots, N$ entry on the map function for kernel $k_2(\underline{x}, \underline{x}')$.

(*) Furthermore, $\phi_k(\underline{x})$ is a $k=1, \dots, M \times N$ vector where the k 'th element is given by

$$\phi_k^{(1)}(\underline{x}) \phi_j^{(2)}(\underline{x}).$$

(note the one-to-one correspondence between each k and an element in the double sum $\sum_{i=1}^M \sum_{j=1}^N$).

Book exercise 6.8)

Suppose that $k_3(x, x') = g(x)^T g(x')$ and let $g(\phi(x)) = f(x)$. Then

$$\begin{aligned} k(x, x') &= k_3(\phi(x), \phi(x')) = g(\phi(x))^T g(\phi(x')) \\ &= f(x)^T f(x'). \end{aligned}$$

Thus, (6.19) defines a valid kernel.

To prove (6.20), let $\underline{x} \in \mathbb{R}^N$ and $\underline{A} \in \mathbb{R}^{N \times N}$ be an $N \times N$ symmetric positive semidefinite matrix. It follows (by some linear algebra result) that $\underline{A}$ can be decomposed as

$$\underline{A} = \underline{Q} \underline{B} \underline{Q}^T,$$

where $\underline{Q} \in \mathbb{R}^{N \times N}$ is an orthogonal matrix and $\underline{B} \in \mathbb{R}^{N \times N}$ is a diagonal matrix with elements ≥ 0 .

Now,

$$\begin{aligned} k(\underline{x}, \underline{x}') &= \underline{x}^T \underline{A} \underline{x}' \\ &= \underline{x}^T \underline{Q} \underline{B} \underline{Q}^T \underline{x}' \\ &= \underbrace{(\underline{Q}^T \underline{x})^T}_{\underline{\gamma}^T} \underline{B} \underbrace{(\underline{Q}^T \underline{x}')}_{\underline{\gamma}'} \\ &= \underline{\gamma}^T \underline{B} \underline{\gamma}' \end{aligned}$$

$$= \sum_{j=1}^N B_{jj} y_j y_j$$

$$= \sum_{j=1}^N (\sqrt{B_{jj}} y_j) (\sqrt{B_{jj}} y_j)$$

$$= \phi(\underline{x})^T \phi(\underline{x})$$

where B_{jj} is the j, j 'th entry on B , y_j is the j 'th entry on $\underline{y}$ (with y_j an analogy to x_j), and $\phi(\underline{x}) \in \mathbb{R}^N$ has entries

$$\sqrt{B_{jj}} y_j = \sqrt{B_{jj}} \underline{Q}^T \underline{x}_j$$

for $j=1, \dots, N$.

Book exercise 6.9)

To prove (6.21), we note that

$$\begin{aligned}k(\underline{x}, \underline{x}') &= k_a(\underline{x}_a, \underline{x}'_a) + k_b(\underline{x}_b, \underline{x}'_b) \\&= \sum_{i=1}^M \phi_i^{(a)}(\underline{x}_a) \phi_i^{(a)}(\underline{x}'_a) + \sum_{j=1}^N \phi_j^{(b)}(\underline{x}_b) \phi_j^{(b)}(\underline{x}'_b) \\&= \sum_{k=1}^{M+N} \phi_k(\underline{x}) \phi_k(\underline{x}') \\&= \phi(\underline{x})^T \phi(\underline{x}')$$

where the k 'th element on $\phi(\underline{x}) \in \mathbb{R}^{M+N}$, for $k=1, 2, \dots, M+N$, is given by

$$\phi_k(\underline{x}) = \begin{cases} \phi_k^{(a)}(\underline{x}_a), & \text{if } 1 \leq k \leq M, \\ \phi_{k-M}^{(b)}(\underline{x}_b), & \text{if } M+1 \leq k \leq M+N. \end{cases}$$

Finally, the proof of (6.22) is analog to that of (6.18).

Book exercise 6.20)

Consider the setup in section 6.4.2.

By (6.64), we know that

$$\underline{t}_{N+1} \sim \mathcal{N}(\underline{t}_{N+1} | \underline{0}, \underline{C}_{N+1}).$$

Rearranged,

$$\underline{t}_{N+1}^* = [\underline{t}_{N+1}, t_1, t_2, \dots, t_N]^T$$

should still follow a Gaussian distribution,
i.e.,

$$\underline{t}_{N+1}^* \sim \mathcal{N}(\underline{t}_{N+1}^* | \underline{0}, \underline{C}_{N+1}^*)$$

where

$$\underline{C}_{N+1}^* = \begin{pmatrix} c & \underline{k}^T \\ \underline{k} & \underline{C}_N \end{pmatrix}$$

with $c = k(\underline{x}_{N+1}, \underline{x}_{N+1}) + \beta^{-1}$ and $\underline{k}$ containing elements $k(\underline{x}_n, \underline{x}_{N+1})$ for $n=1, \dots, N$ as defined in section 6.4.2.

Furthermore, the conditional distribution

$$p(\underline{t}_{N+1} | \underline{t}_N)$$

should also be Gaussian.

Now, in complete analogy to equations (2.94)-(2.98) (treating t_{N+1} as $\underline{x}_a$, $\underline{t}_N$ as $\underline{x}_b$, $\underline{c}$ as $\underline{\Sigma}_{aa}$, $\underline{k}$ as $\underline{\Sigma}_{ba}$, $\underline{k}^T$ as $\underline{\Sigma}_{b,b}$ and $\underline{C}_N$ as $\underline{\Sigma}_{b,b}$), we have from (2.79) and (2.80) that

$$\underline{\Lambda}_{a,a} = (\underline{c} - \underline{k}^T \underline{C}_N^{-1} \underline{k})^{-1}, \quad \text{and}$$

$$\underline{\Lambda}_{a,b} = -(\underline{c} - \underline{k}^T \underline{C}_N^{-1} \underline{k})^{-1} \underline{k}^T \underline{C}_N^{-1}.$$

Substituting into (2.96) and (2.97), we have that

$$p(t_{N+1} | \underline{t}_N) = \mathcal{N}(\underline{\mu}_{a|b}, \underline{\Lambda}_{a,a}^{-1}),$$

with

$$\begin{aligned} \underline{\mu}_{a|b} &= \underline{c} - (\underline{c} - \underline{k}^T \underline{C}_N^{-1} \underline{k}) \cdot (-\underline{k} - \underline{k}^T \underline{C}_N^{-1} \underline{t}_N) \\ &\quad \cdot (\underline{t}_N - \underline{0}) \end{aligned}$$

$$= \underline{k}^T \underline{C}_N^{-1} \underline{t}_N$$

$$= m(\underline{x}_{N+1})$$

and

$$\begin{aligned} \underline{\Lambda}_{a,a}^{-1} &= \underline{c} - \underline{k}^T \underline{C}_N^{-1} \underline{k} \\ &= \sigma^2(\underline{x}_{N+1}) \end{aligned}$$

(note that the mean and covariance matrices become one-dimensional as t_{N+1} is one-dimensional)

Kernel PCA

We want to show that kernel PCA with a linear kernel is equivalent to "standard" PCA.

To this end, we consider the setup in sections 12.1.1 and 12.3. Specifically, the "standard" principal components are defined by the eigenvectors u_d of the covariance matrix

$$S u_d = \lambda_d u_d, \quad S = \frac{1}{N} \sum_{n=1}^N x_n x_n^T = \frac{1}{N} X^T X$$

$D \times 1 \times 1 \times D$ $D \times N \times N \times D$

where the eigenvectors are normalized so $u_d^T u_d = 1$.

- Here, X is a centered data matrix with N rows (i.e. observations) and D columns (i.e. features).

Now, for kernel PCA, we want to solve the problem

$$\frac{1}{N} K a_d = \lambda_d a_d$$

where we define our kernel matrix

$$K = \{k(x_n, x_m)\}_{n,m=1, \dots, N} \quad (12.85)$$

For $k(x_n, x_m) = x_n^T x_m$, we have $K = X X^T$, so

$$\frac{1}{N} X X^T a_d = \lambda_d a_d.$$

The "trick" now, is to realize that

$$S = \frac{1}{N} X^T X \quad \text{and} \quad \frac{1}{N} X X^T$$

have the same eigenvalues, i.e., principal component variances, (up to $n-1$), and that the eigenvectors to $\frac{1}{N} X X^T$ simply are the principal components scaled to have unit variance.

Let v_0 be a unit-length eigenvector to $\frac{1}{N} X X^T$. Then

$$\frac{1}{N} X X^T v_0 = \lambda_0 v_0 \quad (*)$$

$$\Leftrightarrow \frac{1}{N} X^T X (X^T v_0) = \lambda_0 (X^T v_0),$$

thus $X^T v_0$ is a eigenvector of $\frac{1}{N} X^T X$ with eigenvalue λ_0 .

Note that

$$\|X^T v_0\|^2 = (X^T v_0)^T X^T v_0$$

$$= v_0^T X X^T v_0$$

$$\stackrel{(*)}{=} v_0^T (N \lambda_0 v_0) = N \lambda_0 \|v_0\|^2 \stackrel{=1}{=} N \lambda_0$$

and $u_0 = \frac{1}{\sqrt{N \lambda_0}} X^T v_0$ is unit length eigenvector of $X^T X$.